

Velocity Dispersion During the Milky Way Andromeda Merger

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1 INTRODUCTION

This study looks into how velocity dispersion changes throughout the interaction/collision between the Milky Way (MW) and the Andromeda Galaxy (M31). **Velocity Dispersion** is the measure of how randomly stars are moving, a smaller dispersion means stars are moving similarly to one another. These two galaxies are in our **local group**, a collection of many large nearby galaxies as well as the Milky Way’s satellite galaxies. During the MW-M31 merger there are many masses interacting with each other, causing changes in trajectories, which in turn cause velocity changes. Understanding how the velocity dispersion changes during the interaction helps both in predicting the resulting galaxy from the merger as well as understanding how mass is distributed.

Understanding how the velocity distribution changes throughout the collision helps us to understand how galaxies evolve when they interact with their neighbors. A **galaxy** is a gravitationally bound collection of stars whose properties cannot be explained by a combination of baryons and Newton’s laws of gravity. Willman & Strader (2012) Various forces, such as gravity, influence the velocity dispersion, which describes a galaxies evolution whether that be structure, density, or size. **Galaxy evolution** is the process by which a galaxy changes with respect to time, it can be influenced by its stellar population as well as galactic collisions. Developing a better understanding of how velocity dispersion changes within a galaxy during a merger can help us understand how a galaxy might evolve moving forward, and in turn how a galaxy could have evolved to where it is today.

When two galaxies collide, the large disturbance in each of their systems causes a change in stellar movements. The velocity dispersion is a way to measure the movement of stars through a galaxy. When there is a smaller dispersion of velocities in a system it tells us that all the stars are moving in a more ordered pattern, this could explain spiral arms. However, with a greater dispersion the star movement looks more random or chaotic, which is more representative of an elliptical galaxy. When two galaxies merge there are forces that drastically change speeds and directions of where stars are moving, but with each close approach of the two galaxies the stars begin to settle and find a new equilibrium as seen in Figure 1 Stickley & Canalizo (2012). These changes can have an effect on the overall structure of the galaxy as well, for example, a high velocity dispersion supports a thickened disk model Dalcanton et al. (2023).

Many questions on this topic relate to creating more accurate and

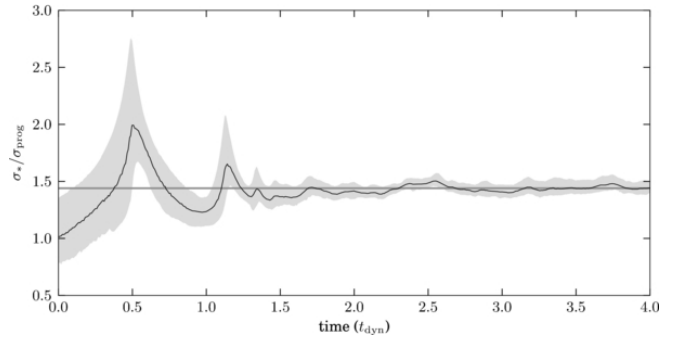


Figure 1. Shows how velocity dispersion changes throughout a merger. Black line is the mean, with uncertainty within the shaded gray region. The gray line is the final mean value of the dispersion.

Stickley & Canalizo (2012)

higher-resolution models to describe and predict velocity dispersion in galaxies Han et al. (2024). To create more accurate models more data is required which is expected of the upcoming completion of the Vera C. Rubin Observatory Pearson, W. J. et al. (2024). Following more accurate models higher resolution models can be constructed to consider details such as dust absorption as well as 3D velocity dispersion.

2 THIS PROJECT

In this paper, we will evaluate how the velocity dispersion of each galaxy evolves throughout the MW/M31 interaction. At various important steps during the interaction we will look into how the velocity dispersion changes throughout the galaxy up to its Jacobi Radius. The **Jacobi Radius** is the radius at which a particle is no longer gravitationally bound to a system. Beyond that radius, particles are no longer bound by a galaxy’s gravity. Furthermore, the interesting points during the interaction will be when the two centers of mass are at a local maxima or minima. These points of interest can be compared to the initial velocity dispersion to draw conclusions on galactic evolution throughout the merger.

This project addresses trying to predict galaxy evolution through velocity dispersion with high-resolution models of the merging event.

This question is important to look into because it provides the opportunity to deepen our understanding of galaxy evolution. As we

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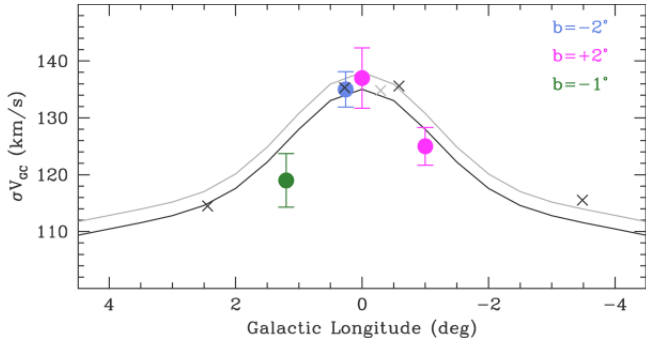


Figure 2. Shows how velocity dispersion changes along the longitude of the Milky Way. The solid lines are the expected radial velocity dispersion trends. While the individual points represent observations at various galactic latitude (b).

Pearson, W. J. et al. (2024)

currently understand them, galactic collisions are a driver for stellar evolution. [Schawinski et al. \(2010\)](#) It can be difficult to measure velocity dispersion of other galaxies due to limited resolution, thus high-resolution simulations can help us draw connections between the snapshots of galaxies we see today. This study will help by analyzing high-resolution simulation to present the evolution of velocity dispersion throughout the MW/M31 interaction.

3 METHODOLOGY

This simulation utilized in this study is an N-body smoothed-particle hydrodynamics code, GADGET3. [Springel \(2005\)](#) An N-body simulation is a simulation used to model the motion of N individual particles and their gravitational influence on one another. The model incorporates star formation and supernova feedback which is not typically used in other cosmological simulations. [Springel & Hernquist \(2003\)](#) Each galaxy in this simulation is modeled by three components disk, bulge, and halo. The **stellar disk** is the flat component of the galaxy where spiral arms may form and young stars are born. The **stellar bulge** is the bright central region of a galaxy which is usually spherical. The mass of the halo component is inferred based on the total observed mass of the galaxy which is used to define the Hernquist dark matter profile. [Besla et al. \(2012\)](#)

For this study, first, we utilize the low-resolution models produced to develop initial results. Following that, higher resolution models will be used in order to generate more impactful results.

The most important calculation in this question is the velocity dispersion σ . It can be calculated just like a regular standard deviation, first, calculating the mean velocity and then calculating the squared average deviation from the mean in each coordinate direction:

$$V_{mean} = \frac{1}{N} \sum_{i=1}^N v_i \quad (1)$$

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (v_i - V_{mean})^2} \quad (2)$$

Where N is the number of particles in one measurement and v_i is the velocity of the i th particle. The code will calculate the velocity dispersion of shells of material at constant radii moving outwards

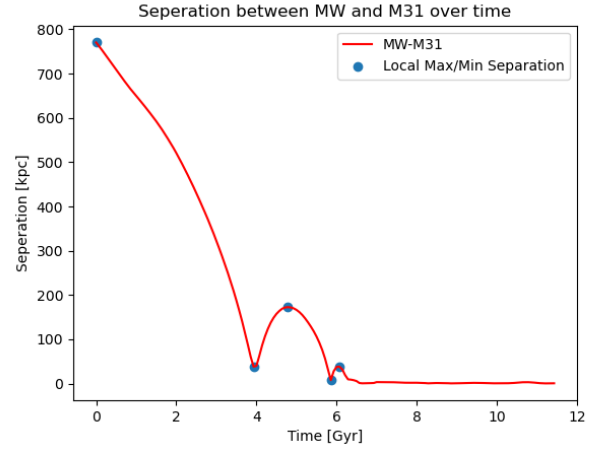


Figure 3. Separation between MW and M31 throughout their interaction. The red line shows the separation between the two galaxies over time. The blue dots depict local maxima and minima throughout their interaction which are points of interest as described in the 'This Project' section. The y-axis shows the separation in Kpc and the x-axis displays the time during the simulation in Gyr with 0 being present day.

until the Jacobi Radius (R_J). [King \(1962\)](#)

$$R_J = R \left(\frac{m}{2M} \right)^{1/3} \quad (3)$$

Where R is the separation between the two interacting bodies, m is the mass of the satellite galaxy, and M is the total mass in the system. Only Bulge and Disk type particles will be considered because there isn't an observational way to verify the movement of dark matter particles. The code first slices the galaxy into many spherical shells. Within each of these shells eq. 1 will be used to calculate the mean velocity within of bulge and disk particles. Following that eq. 2 will be used to calculate the velocities dispersion within that shell. This process will be repeated for each minima or maxima for the separation between MW and M31.

To answer these questions I will need two distinct plots. The first plot simply shows the separation between MW and M31 highlighting the maxima and minima throughout their interaction. Following that, at each of the minima or maxima I will need to produce a plot that displays the velocity dispersion as a function of radius similar to what is done in Figure 2. [Pearson, W. J. et al. \(2024\)](#) By developing an understanding of how the velocity dispersion changes radially through time we can understand how the shape of a galaxy would change as well.

Based on the work of Stickley et al. (2012), it is suggested by his N-body simulations that the velocity dispersion increases when a galaxy merger occurs. As Figure 1 shows the final equilibrium of the velocity dispersion increases from the value at $t = 0$. I predict that this is happening because of the addition of mass and energy into a system bringing about chaos as the stars attempt to find a new equilibrium. I suspect that following the interaction between the two galaxies the result will be elliptical as the velocity dispersion increases as a whole.

4 RESULTS

As shown in Figure 3 MW and M31 have multiple close approaches before they merge around 6.5 Gyr into the simulation. Each of these

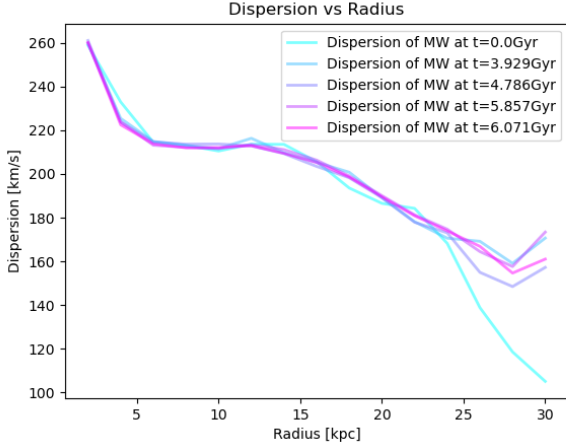


Figure 4. Depicts the velocity dispersion as it changes radially outward from the center of MW. Each continuous line shows how the dispersion changes radially at different times throughout the simulation. The lightest blue represents 0Gyr (today) and as it gets more pink the age progresses to 6Gyr. The y-axis shows the Velocity Dispersion in Km/s. The x-axis show the radius in units of Kpc and extends until 30 Kpc which is treated as the edge of MW.

approaches provide insight into how galaxies evolve when they are disturbed. This figure highlights the important snapshots of the simulation that are the focus of this study. The separation is important to understand why the MW particles might accelerate or tend to a certain direction which ultimately affects the dispersion. This figure highlights two maxima, two minima and a baseline separation between MW and M31 and serves as a building block for the rest of the study.

In figure 4 we produce interesting results. This is especially visible on the farther end of the radius axis. In the present day line we can see that the velocity dispersion reaches lower values whereas after the first close approach of the two galaxies the velocity dispersion is higher. However, what is even more interesting is that their interaction before a minima and before a maxima are also different. For a line that represents a minima the line increases from the snapshot that precedes it, and a line representing a maxima slightly decreases from the snapshot that precedes it. Additionally, from one minima to the next minima and from one maxima to the next maxima the velocity dispersion increases. So not only is velocity dispersion changing at each of the interesting snapshots but the velocity dispersion as a whole is trending upwards throughout their interaction even if it decreases at times. Considering only the difference between the first and last snapshot (0-6.071 Gyr) the difference in the dispersion at farther radii (24-30 kpc) is in the range of 5-56 km/s, depending on the radius. At farther radii the velocity dispersion throughout the interaction increases adding to the disorder and chaotic order of particles (or stars) in these regions.

5 DISCUSSION

The velocity dispersion does tend to increase at larger radii. This agrees with the hypothesis that as the two galaxies interact there should be an increase in the velocity dispersion of the galaxy. The velocity dispersion calculated at the farthest radii (24-30 kpc) increase from 5.7 km/s to 55.9 km/s. This is a significant increase in

velocity dispersion for the particles that are farther from the center of MW.

The increase of velocity dispersion depicted in figure 4 was similarly shown in figure 1. [Stickley & Canalizo \(2012\)](#) Additionally, the oscillatory motion of the velocity dispersion as it settles into a new equilibrium is also shown in figure 1. [Stickley & Canalizo \(2012\)](#) These results further verify that the motion of stars within a galaxy are greatly perturbed throughout a galactic merger. However, it is also interesting to see that the oscillatory changes in the velocity dispersion suggest the particles finding a new equilibrium as the two galaxies become one.

In the analysis of the data the biggest source of error likely came from the calculation of velocity dispersion. There were two methods used to calculate dispersion which resulted in two different results. The first method calculates the 3D velocity and then finds dispersion of that velocity and the second method calculate the dispersion of each component of the velocity and then finds the magnitude of the dispersion. The second method was utilized in this study. An additional source of error comes from neglecting the effects of M33 which is also present in the simulation. Because M33 has a gravitational influence on the particles in the simulation it is causing some change in the velocities of the particles in MW.

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